Sustainable Heating and Cooling Technologies
Spring Semester 2026
Student Project
29.04.2026

Student Project: Cooling a server room

Effective cooling is essential for any data center, as servers generate heat that can lead to system failure if not managed properly. As an engineer, your task is to ensure the hardware stays within safe operational limits to prevent any server downtime.

In this project, you are responsible for designing and dimensioning a cooling system for the air in a server room that should be kept at around 15°C. The server room is unaffected by external heat sources such as solar radiation or heat transfer through the walls. The only heat source is the servers. To meet the required conditions, you have two options: **ambient air ventilation** and **active air conditioning (AC)**. The cooling system can switch between these two cooling modes. Your goal is to dimension the system to keep the server room's air temperature within a target temperature range over the course of a year, as represented by four example days.

Think critically about which aspects are worth considering and which assumptions can be made to simplify the evaluation. The aim is to achieve an acceptable design for the server room.

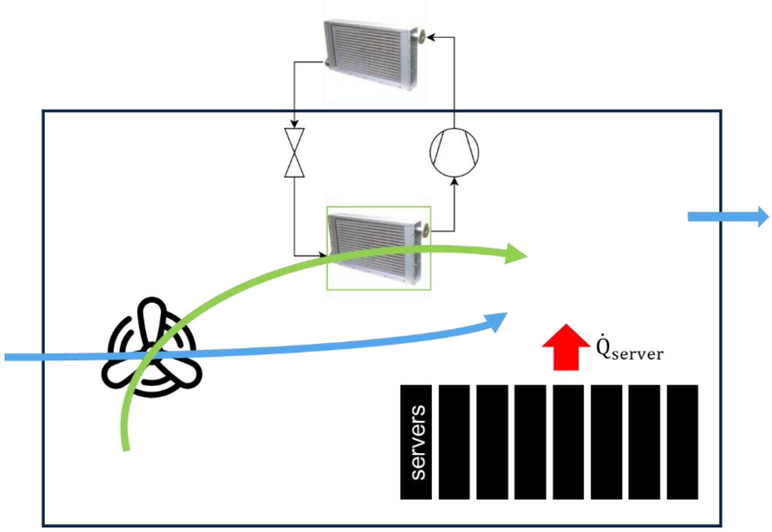

Figure 1 System for server room cooling

Sustainable Heating and Cooling Technologies
Spring Semester 2026
Student Project
29.04.2026

Your task:

1. Determine the required specifications of the AC unit
    - Estimate the required cooling power of the AC unit.
    - Estimate volumetric flow limits for the system.
    - Develop an on/off control strategy for the AC and ventilation system that keeps the server rooms' air temperature within an acceptable range.
2. Design the thermodynamic cycle of the AC unit to meet the requirements of the server room
    - Design the AC cycle as a function of the given compressor size and refrigerant type.
    - Take into account typical technical constraints.
3. Simulate the operation of the server room cooling system, including the AC cycle and the ventilation, for the given combinations of compressor size and refrigerant and select the most suitable combination of compressor size and refrigerant
    - Consider the four example days of a year
    - When selecting the best design, consider overall energy demand, the number of start/stop cycles of the AC compressor, operating times, and the server room air condition (temperature, humidity)
    - Visualize the operation of the complete system (room air, AC, ventilation) over all four days using appropriate plots.
4. Estimate the electricity cost of the air conditioning and the ventilation system, and discuss your results

Design requirements

System modeling:

Parameters:

- "ambient_temperature_fall/winter/summer/spring.txt" list the ambient air temperature of the environment during four typical days per year. The relative humidity is assumed to be constant at 60%.
- "server_heating_power.txt" indicates the time-dependent heat flow rate resulting from server operation.
- For simulation of the cooling system's operation, a temporal resolution of 5 min is sufficiently accurate.
- The system can be considered in quasi-steady-state.

- The system is not equipped with an additional active (de-)humidifier.

Server room:

- The server room has the following dimensions: $l_{\mathrm{i}} = 10\ \mathrm{m}$, $h_{\mathrm{i}} = 3\ \mathrm{m}$, $w_{\mathrm{i}} = 6\ \mathrm{m}$ (inside)
- The volume of the server can be neglected, and the mass of dry air in the room is assumed to be constant.
- The air in the server room can be modeled as 0-d.
- The initial room air temperature is 15°C and the initial relative humidity is 60%.

Air conditioning unit:

Process

- Consider a subcritical vapor compression cycle.
- Heat source: Server room air (re-circulation)
- Heat sink: Ambient air
- Implement realistic technical constraints (e.g., superheating) and ensure the system is in COP-optimal operation at each point in time.
- To maintain the compressor's operating envelope, consider a minimum pressure ratio of 2.
- Assuming constant approach temperatures is reasonable, also in off-design operation.

Refrigerant

Investigate the following refrigerants: Propane, R1234yf, Dimethyl ether

Compressor

- Use the function **`recip_comp_SP`** to determine the isentropic compressor efficiency and the refrigerant mass flow (see description below).
- The compressor model considers a 4-cylinder reciprocating compressor that is available in different cylinder diameters (bore) D = {30 mm, 40 mm, 50 mm}.

Ventilation:

- Consider acoustic and flow velocity constraints that might limit the volume flow.

Controlling:

- Ventilation of ambient air and the air flow through the AC's evaporator rely on the same ventilator.

- The AC unit and the ventilation are on/off controlled. Implement a system control logic that determines whether ventilation with ambient air or the AC unit is used to cool the server room air.
- To decide whether the AC unit shall be switched on/off, the controller relies on temperature sensors exclusively, and you need to consider threshold values.
- Typical minimal standstill and running times of air conditioning units are 10 and 5 min.
- Choose the cooling power of the AC unit and the ventilation wisely to avoid undesirable temperature fluctuations.

Hint :

1. During simulation of the cooling system's operation, optimizing the AC's thermodynamic cycle is time-consuming, especially when simulating an entire day. Thus, we recommend creating a function that approximates the air-conditioning COP_{inner} and cooling capacity over a defined range of source/sink temperatures (off-design), based on detailed process optimization. Instead of calling the optimization routine for each timestep, you can use the data resulting from this function to represent the AC's performance in your simulation, e.g., through interpolation.
2. The AC compressor is on/off controlled. The COP degrades if the AC unit delivers a higher cooling power than requiredfor cooling the server room. The resulting COP_{res} can by calculated as function of the inner COP_{inner} through the following empirical equation:

$$\mathrm{COP_{res}} = \mathrm{COP_{inner}} \left(\frac{\frac{\dot{Q}_{\mathrm{demand}}}{\dot{Q}_{\mathrm{AC}}}}{0.9\,\frac{\dot{Q}_{\mathrm{demand}}}{\dot{Q}_{\mathrm{AC}}} + 0.1} \right)$$

`recip_comp_SP(param, refrigerant, transcrit=False)`

Estimates the **isentropic efficiency** and **mass flow rate** of a **reciprocating compressor** under **subcritical conditions** using empirical correlations.

Parameters

- **`param`** (tuple):
  Input parameters as a 5-element list:
  (T_ev, T_co, DeltaT_sh, DeltaT_sc, D)
  - T_ev (float): Evaporating temperature [°C]
  - T_co (float): Condensing temperature [°C]
  - DeltaT_sh (float): Superheat at suction [K]
  - DeltaT_sc (float): Subcooling before throttle [K]
  - D (float): Cylinder diameter [mm]
- **`refrigerant`** (str):
  Refrigerant name (Coolprop).
- **`transcrit`** (bool, optional):
  Indicates whether the operation is transcritical. Should be False for subcritical use (default).

Returns

- `eta_is` (float): Isentropic efficiency (dimensionless)
- `m_dot` (float): Mass flow rate [kg/s]
